ASMOE V2: Heterogeneous Mixture-of-Experts with Adaptive Soft Routing for Potato Leaf Disease Classification in Uncontrolled Environments

Long Vũ Trần[1*], Hoàng Anh Kiệt Nguyễn[1] and Ngọc Thành Nhân Nguyễn[1]

[1*]FPT University, Da Nang, Vietnam.

*Corresponding author(s). E-mail(s): vutran.03092006@gmail.com;
Contributing authors: kiet.nguyen20067@gmail.com;
nnmtreallove@gmail.com;

Abstract

Automatic diagnosis of potato leaf diseases from field photographs is hampered by cluttered backgrounds, an 11-fold class imbalance and visually similar symptoms of insect and fungal damage. Auditing an earlier experimental campaign on the 3,076-image Potato Leaf Disease Dataset in Uncontrolled Environment, we found 55 byte-identical file groups (six of them carrying conflicting labels, all involving the Pest class) and 27 near-duplicate bursts. When folds were drawn at random, these duplicates inflated cross-validated Macro-F1 from 83.68% to 88.16%. We propose ASMOE V2, an adaptive soft mixture of heterogeneous experts (DenseNet-201, EfficientNetV2-M and Swin-B) whose feature maps are aligned by a LayerNorm and $\mathbf{1} \times \mathbf{1}$ convolution projection and fused by an input-conditioned soft router. The model is trained with a two-stage refinement schedule, discriminative learning rates, square-root class weighting and a supervised contrastive loss, on leaf-cropped inputs at up to $\mathbf{512} \times \mathbf{512}$ pixels. Under a leakage-free, group-aware stratified 5-fold protocol, ASMOE V2 reaches $92.63 \pm 0.91\%$ accuracy and $91.75 \pm 1.88\%$ Macro-F1 ($+8.07$ pp over the strongest legacy model), with Nematode F1 of 89.0%. Pest/Fungi confusions fall by about two thirds, while Phytophthora/Fungi remains the main error source. The learned routing is strongly class-dependent, and routing-weighted Grad-CAM shows how the experts divide the evidence among themselves.

Keywords: Potato leaf disease, Mixture-of-Experts, Adaptive soft routing, Class imbalance, Data leakage, Grad-CAM

1 Introduction

Potato is one of the most widely consumed staple crops, and foliar diseases caused by bacteria, fungi, oomycetes, viruses, nematodes and insect pests are a major source of yield loss. Early, reliable identification from leaf images taken with ordinary phones would let growers act before symptoms spread. Field photographs, however, differ sharply from the curated, plain-background images on which most leaf-disease classifiers are benchmarked. Leaves are photographed among soil, hands and neighbouring plants, under changing illumination, at varying distances. Several disease classes produce similar necrotic or chlorotic lesions.

The Potato Leaf Disease Dataset in Uncontrolled Environment [1] captures these conditions in 3,076 images of seven classes collected in Central Java, Indonesia. Transfer learning with single CNN backbones [1] and hybrid CNN–Transformer designs [2] have been evaluated on it. Recently, Hoang and Phan [3] proposed ASMoE, an adaptive soft mixture of EfficientNet-B0, DenseNet-121 and Swin-Tiny experts whose input-dependent gate weights the three architectures per image. On a single stratified 81/9/10 split they reported 92.62% F1.

Two questions remain open. First, how much of the reported performance on this dataset reflects generalisation, given that the collection contains duplicated and near-duplicated photographs? Second, can the mixture-of-experts idea be scaled, with larger heterogeneous experts, higher input resolution and a training recipe aimed at the minority classes, without overfitting 3,000 images?

This paper answers both. Our contributions are:

1. A **forensic audit** of an earlier campaign of eight models (Section 2). It quantifies label-conflicting and redundant duplicates, the leakage they cause under random splits, the confusion structure among Pest, Fungi and Phytophthora, the failure of CutMix, and the under-training of the earlier models.
2. A **leakage-free data protocol**. Conflicting duplicates are removed, redundant copies deduplicated, and near-duplicates grouped by perceptual hashing. Folds are drawn with stratified group k-fold, and leaf regions are cropped by a foliage-and-lesion colour mask.
3. **ASMOE V2**, a heterogeneous mixture (DenseNet-201, EfficientNetV2-M, Swin-B) with a feature projection module, an input-adaptive soft router, multi-resolution experts and a two-stage refinement schedule with supervised contrastive learning (Section 3).
4. A **reproducible evaluation** against the legacy models on identical group-aware folds, with per-class and confusion analysis, routing statistics and routing-weighted Grad-CAM. All numbers in this manuscript are regenerated automatically from the experiment logs (Section 4).

2 Dataset and Baseline Audit

Before designing a new model we audited every artefact left by the earlier experimental campaign: four Kaggle notebooks trained under a plain stratified split (EfficientNet-B0, MobileViT-S, a sequential EfficientNet→MobileViT cascade and an EfficientNet→Swin cascade) and a later, optimised campaign that introduced a group-aware split, a policy screen, an architecture screen and a final 5-fold cross-validation (CV). All metrics, per-epoch logs, confusion matrices and duplicate reports were parsed directly from the archives; model checkpoints were not needed for the analysis.

2.1 Dataset composition and class imbalance

The Potato Leaf Disease Dataset in Uncontrolled Environment [1] contains 3,076 field photographs in seven classes (Table 1). The distribution is strongly skewed: *Fungi* has 748 images while *Nematode* has only 68, an imbalance ratio of 11.0×; *Healthy* (201) is the second-smallest class. With five folds, each validation split contains only 11–17 Nematode images, so a single misclassified sample moves Nematode recall by 6–9 percentage points. This explains the large fold-to-fold variance of minority-class metrics (Nematode F1 SD = 4.0 pp for the selected legacy model) and motivates reporting Macro-F1 alongside accuracy.

Table 1 Class distribution, redundant byte-level copies and post-cleaning counts.

Class	Images	Share (%)	Redundant copies	After cleaning
Bacteria	569	18.5	0	569
Fungi	748	24.3	5	740
Healthy	201	6.5	1	199
Nematode	68	2.2	0	67
Pest	611	19.9	4	601
Phytophthora	347	11.3	38	309
Virus	532	17.3	1	530
Total	3076	100.0	49	3015

Class weighting was the single most effective intervention in the legacy policy screen (Table 3): square-root inverse-frequency weights raised fold-1 Macro-F1 from 70.89 % to 74.65 % and Nematode recall from 34.4 % to 81.3 %. However, the final confusion matrices show the cost of this re-weighting: pooled over five folds, Nematode recall reached 89.7 % but precision only 67.0 %, because 14 Fungi and 9 Pest images were pushed into the minority class. Healthy shows the same pattern (recall 86.1 %, precision 75.2 %), absorbing 27 Pest and 20 Virus images. The minority classes therefore suffer from *false positives*, not missed detections, which loss re-weighting alone cannot fix.

2.2 Byte-level and near-duplicate contamination

SHA-256 hashing found 110 files forming 55 groups of byte-identical images. Six of these groups carry *conflicting labels*: one Virus/Pest pair, one Pest/Healthy pair, three

Pest/Fungi pairs and one Pest/Nematode pair. Every conflict involves the Pest class, which is also the class with the lowest recall (75.6 %) and the dominant source of confusion with Fungi (Section 2.3). The same files are, in effect, annotated as two diseases, so a perfect classifier cannot score 100 % on them, and if the two copies are split across train and validation the model is penalised for reproducing a training label.

The other 49 groups contain the same label twice. Thirty-eight of these redundant copies belong to *Phytophthora* and look like download artefacts (`IMG_9518.JPG` vs. `IMG_9518 (1).JPG`). A 64-bit difference hash (dHash) found a further 27 near-duplicate pairs with Hamming distance ≤ 2 (Fungi 9, Virus 7, Phytophthora 5, Bacteria 4, Pest 2), mostly consecutive burst shots (`IMG_1111` / `IMG_1112`). None of them crosses classes.

The original four notebooks used `StratifiedKFold` on raw file paths, so any duplicate or burst pair could appear in both the training and the validation split. Their reported scores (Table 2, "Random") are therefore optimistically biased. Once the optimised campaign moved to a group-aware split, MobileViT-S fell from 88.16 % to 83.68 ± 1.42 % Macro-F1. That drop should not be attributed to leakage alone: the two campaigns also differ in learning rate (full fine-tuning at 10^{-3} vs. 2×10^{-5} for the backbone and 5×10^{-4} for the head) and in training length (15 fixed epochs vs. an early-stopped schedule capped at 14). We treat the group-aware number as the only valid baseline.

Table 2 Historical 5-fold CV results recovered from the legacy Kaggle archives. Random splits ignore duplicate groups and are therefore optimistically biased.

Model	Split protocol	Res.	Accuracy (%)	Macro-F1 (%)	MCC
MobileViT-S	Random (leaky)	224	89.89 ± 1.69	88.16 ± 2.68	0.877 ± 0.021
EfficientNet-B0	Random (leaky)	224	86.87 ± 1.99	85.03 ± 3.27	0.840 ± 0.024
EffNet-B0→MobileViT	Random (leaky)	224	86.09 ± 1.31	84.60 ± 1.98	0.831 ± 0.016
EffNet-B0+MobileViT (ECA)	Group-aware	224	83.55 ± 2.20	82.12 ± 2.67	0.801 ± 0.027
MobileViT-S	Group-aware	224	84.92 ± 1.27	83.68 ± 1.42	0.817 ± 0.016

2.3 Confusion patterns

Figure 1 shows the pooled 5-fold confusion matrix of the selected legacy model (MobileViT-S, sqrt class weights, no augmentation). Errors are concentrated in a small number of biologically plausible pairs:

- **Pest ↔ Fungi** is the largest error mode in both directions: 76 Pest images predicted as Fungi (12.4 % of Pest) and 69 Fungi images predicted as Pest (9.2 % of Fungi). Insect feeding damage and fungal necrosis both produce irregular brown lesions with chlorotic halos, and at 224×224 the fine cues that separate them (frass, feeding holes and leaf margins vs. concentric rings and sporulation texture) are only a few pixels wide. Three of the six label-conflicting duplicates are exactly Pest/Fungi pairs, so part of this confusion is label noise in the data.

- **Pest → Virus / Healthy** (26 and 27 images): mild feeding damage on otherwise green leaves resembles both mosaic-type viral symptoms and healthy foliage.
- **Phytophthora ↔ Fungi** (31 and 26 images): late blight lesions are themselves oomycete-induced necroses that look like other fungal leaf spots.
- **Virus → Healthy** (20 images): early mosaic and leaf-roll symptoms are subtle colour changes that the ImageNet colour statistics and 224 px inputs do not preserve well.

The two-branch EfficientNet-B0 + MobileViT ECA-gated fusion reproduced the same pattern (Pest→Fungi 64, Fungi→Pest 66) with higher variance ($82.12 \pm 2.67\,\%$ Macro-F1). Adding a second backbone of a similar scale therefore did not provide complementary evidence for the hard pairs.

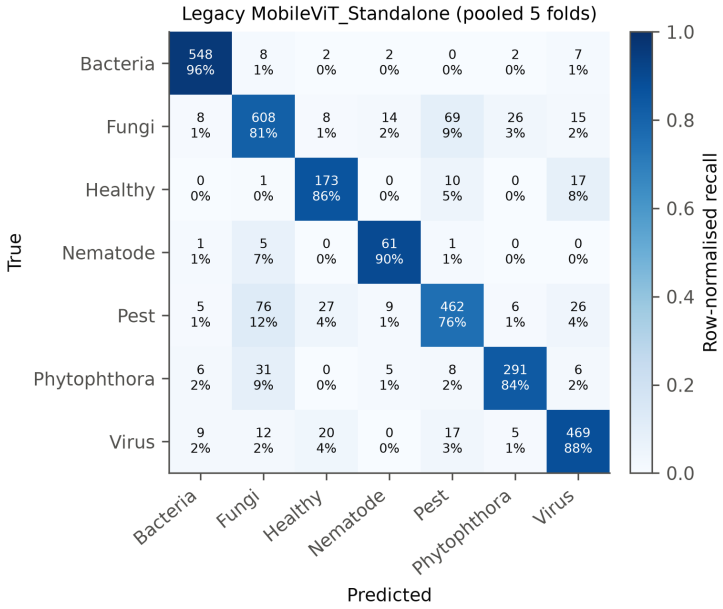


Fig. 1 Pooled 5-fold confusion matrix of the selected legacy model (MobileViT-S, group-aware split). Colour encodes row-normalised recall; cells show counts and percentages.

2.4 Why aggressive CutMix degraded Macro-F1

In the policy screen (Table 3) CutMix produced the lowest Macro-F1 of all policies, 57.89 %, against 70.89 % without augmentation at the same class weighting. It also nearly eliminated minority-class detection (Nematode recall 3.1 %, Pest recall 30.0 %). We attribute this to the mismatch between CutMix's assumptions and the structure of leaf-disease images:

1. **Label/evidence misalignment.** CutMix mixes labels in proportion to the pasted *area*, but the discriminative evidence is a few small lesions that cover a small

Table 3 Legacy augmentation/class-weight screen (MobileViT-S, fold 1, 4 epochs).

Augmentation	Class weight	Macro-F1	Weighted-F1	Nematode rec.	Pest rec.
none	sqrt	74.65	74.95	81.2	45.7
none	none	70.89	74.32	34.4	46.5
mixup	none	69.27	73.08	31.2	43.0
cutmix	none	57.89	66.68	3.1	30.0

fraction of the leaf. A random box removes the lesion in many samples, or pastes a lesion-free patch, while the target still assigns 30–70 % probability to the original disease. For seven classes that share the same green leaf background, this injects heavy label noise.

2. **Destroyed lesion boundaries.** Lesion margins, halos and concentric rings are what separate Pest from Fungi and Phytophthora. The sharp rectangular seams introduced by CutMix create artificial edges at exactly the spatial frequencies the network should learn from.
3. **Minority starvation.** With only 52–57 Nematode images per training fold, the few minority samples are often pasted over or used as fillers, so their clean, full-label gradient signal becomes even rarer.
4. **Slower convergence.** The screen ran for only four epochs; CutMix is known to need longer schedules. Part of the gap is therefore an under-training effect, but the direction is consistent with points 1–3. MixUp lost a smaller 1.6 pp, which is consistent with the view that globally blending two whole leaves keeps lesion morphology intact while CutMix does not.

These observations motivate the augmentation policy of ASMOE V2: geometric transforms that preserve lesions (rotation $\pm 20°$, mild affine, flips), mild photometric jitter (brightness and contrast 0.15), and *mild* MixUp ($\alpha = 0.2$, $\lambda \geq 0.5$, applied to half of the batches), with CutMix removed entirely.

2.5 Training-budget bottleneck

The learning curves of the final legacy CV (Fig. 2) show that the selected model was still improving when training stopped: the best epochs were 12, 14, 13, 13 and 9 under a 14-epoch cap, and the training loss ($\approx$ 0.37–0.40) was still close to the validation loss ($\approx$ 0.41–0.50). The screening stages used only four epochs, so the policy and architecture decisions were taken on under-trained models. ASMOE V2 therefore uses a two-stage schedule with up to 45 epochs and patience-based early stopping on Macro-F1.

2.6 Summary of bottlenecks

1. Class imbalance of 11$\times$ with re-weighting-induced false positives for Nematode and Healthy.
2. Label noise: 6 cross-class byte duplicates (all involving Pest), 49 redundant copies, 27 near-duplicate bursts; leakage under random splits.

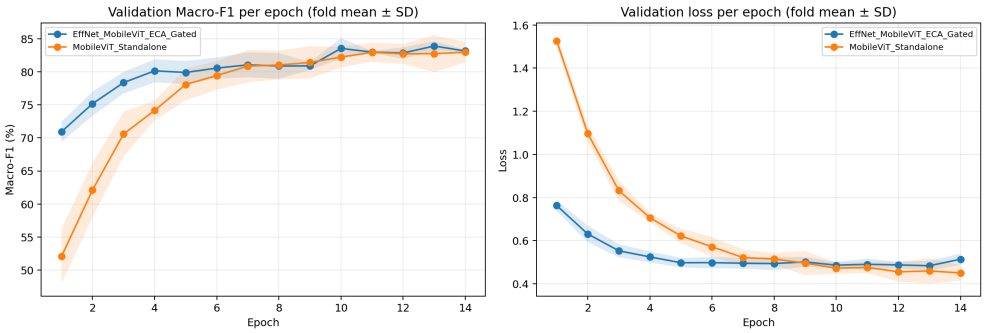


Fig. 2 Legacy final-CV learning curves (fold mean ± SD). Both models were still improving when the epoch cap was reached.

3. Fine-grained Pest/Fungi/Phytophthora confusion that 224 px inputs cannot resolve.
4. Augmentation that destroys lesion evidence (CutMix).
5. Under-training (best epoch at or near the 14-epoch cap) and screening decisions taken after only four epochs.
6. Homogeneous, lightweight backbones (4.9–10.5 M parameters) with no global-context expert.

3 Methods: ASMOE V2

ASMOE V2 targets the bottlenecks identified in Section 2 along three axes: *data* (cleaning, leakage-free folds, leaf-centred inputs at higher resolution), *architecture* (heterogeneous experts aligned in a shared space and fused by input-adaptive soft routing), and *optimisation* (two-stage refinement with discriminative learning rates and a supervised contrastive objective). Figure 3 gives an overview.

3.1 Data curation and leakage-free cross-validation

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$ be the raw collection. We compute the SHA-256 digest s_i and a 64-bit difference hash d_i of each file, and apply three rules: (i) every exact-duplicate group $\{i : s_i = s\}$ whose labels disagree is removed entirely, since its ground truth is undefined; (ii) within label-consistent exact-duplicate groups only one copy is kept; (iii) surviving images with $\mathrm{Ham}(d_i, d_j) \le \tau$ ($\tau = 2$) are joined by union-find into one *group* g_i. Cross-label near-duplicates, if any, are removed like rule (i). Folds are produced by `StratifiedGroupKFold`(K=5), which keeps class proportions while guaranteeing $g_i = g_j \Rightarrow \mathrm{fold}(i) = \mathrm{fold}(j)$. The code checks this invariant explicitly. All models in this paper, including the re-trained baselines, share one fixed split.

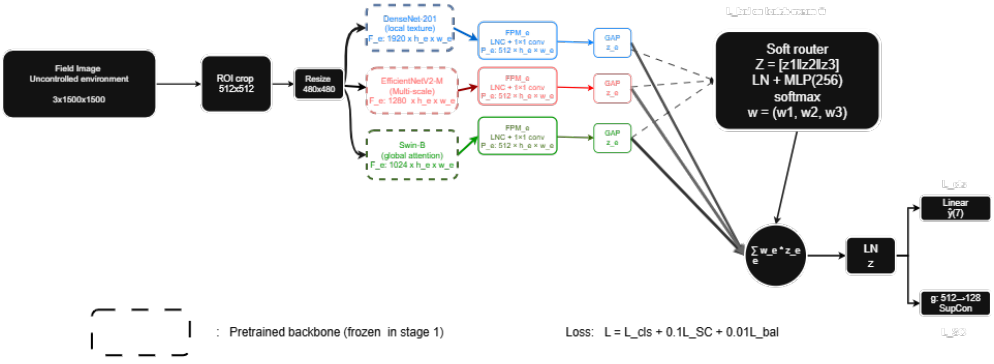


Fig. 3 ASMOE V2 overview. Each expert sees the image at its native resolution; the Feature Projection Module (FPM) aligns heterogeneous feature maps; the soft router produces per-image expert weights.

Leaf ROI cropping.

Field photographs contain soil, hands, sky and neighbouring plants. We build a foliage-and-lesion mask

$$M = \underbrace{[\,2G - R - B > 0.05\,]}_{\text{excess green}} \vee [\,H \in [35°, 170°],\, S > 0.18\,] \vee \underbrace{[\,H \in [5°, 35°),\, S > 0.25\,]}_{\text{chlorosis / necrosis}}, \tag{1}$$

gated by $V > 0.12$, smoothed with a 5×5 median filter. We take the 2nd–98th percentile extent of M, add an 8 % margin and square it. Including yellow-brown hues prevents necrotic lesions, the most informative pixels, from being cropped away. If the mask covers less than 5 % of the image, or the box would cover less than 15 % of it, the full frame is kept. Crops are cached once at 640 px on the longer side.

Resolution.

Inputs are loaded at 512×512 (vs. 224 in all legacy runs), and each expert resamples them to its pre-training resolution (512, 480, 384). At 512 px a 2 mm lesion on a 10 cm leaflet spans ≈10 pixels instead of ≈4.

Lesion-preserving augmentation.

Training uses horizontal and vertical flips, rotation $\mathcal{U}(-20°, 20°)$, affine translation (±5 %), scale ($[0.9, 1.1]$) and shear (±5°), and ColorJitter (brightness and contrast 0.15). CutMix [4] is removed (Section 2.4). On half of the mini-batches we apply mild MixUp [5] with $\lambda \sim \text{Beta}(0.2, 0.2)$ and $\lambda \leftarrow \max(\lambda, 1-\lambda)$, so the dominant image always contributes at least half of the pixels and the label:

$$\tilde{x} = \lambda x_a + (1-\lambda) x_b, \qquad \tilde{q} = \lambda\, q_a + (1-\lambda)\, q_b, \tag{2}$$

where q are label-smoothed one-hot targets ($\epsilon = 0.05$).

3.2 Heterogeneous experts

Each expert E_e is chosen for a different inductive bias that matches one of the observed confusions:

- **Local texture, DenseNet-201** [6] (or ConvNeXt-B in the large variant). Dense connectivity reuses low-level features in every block, so fine texture such as sporulation, frass and lesion rims, which separates Pest from Fungi, reaches the classifier without being washed out by depth.
- **Multi-scale, EfficientNetV2-M/L** [7]. Compound scaling and Fused-MBConv stages provide receptive fields from a few pixels to the whole leaf. This matters because lesion size varies by an order of magnitude between early and late infections.
- **Global attention, Swin-B/L** [8]. Shifted-window self-attention relates distant regions, such as a mosaic pattern spread over the leaf or leaf-roll curvature. This is the evidence needed for Virus vs. Healthy, where no single local patch is diagnostic.

All experts are initialised from ImageNet-21k/1k weights (timm). For an input x, expert e resamples it to its own resolution r_e and returns a feature map $F_e = E_e(\text{resize}(x, r_e)) \in \mathbb{R}^{C_e \times h_e \times w_e}$, with $C_e \in \{1920, 1280, 1024\}$. Token outputs of the transformer are reshaped to a spatial map.

3.3 Feature Projection Module

The experts differ in channel count, spatial size and activation statistics (BatchNorm-ReLU vs. LayerNorm-GELU). Before any fusion, the FPM maps each of them into a shared D-dimensional space ($D = 512$):

$$P_e = W_e * \mathrm{LN}_C(F_e) + b_e, \qquad \mathrm{LN}_C(F)_{:,u,v} = \gamma \odot \frac{F_{:,u,v} - \mu_{u,v}}{\sigma_{u,v}} + \beta, \tag{3}$$

where LN_C normalises across channels at every spatial location and $W_e \in \mathbb{R}^{D \times C_e \times 1 \times 1}$ is a pointwise convolution. Per-location normalisation removes the scale mismatch between CNN and transformer features while keeping the spatial layout, which Grad-CAM later uses. The expert descriptor is

$$z_e = \frac{1}{h_e w_e} \sum_{u,v} P_e[:, u, v] \in \mathbb{R}^D. \tag{4}$$

3.4 Input-adaptive soft routing

Let $Z = [z_1 \| \dots \| z_E] \in \mathbb{R}^{ED}$. The gating network is a two-layer MLP followed by a temperature-scaled softmax:

$$w = \mathrm{softmax}\left(\frac{W_2\, \delta\big(W_1\, \mathrm{LN}(Z) + b_1\big) + b_2}{T} \right) \in \Delta^{E-1}, \tag{5}$$

with δ the GELU, $W_1 \in \mathbb{R}^{256 \times ED}$, dropout 0.1 and $T = 1$. W_2 and b_2 are zero-initialised, so training starts from uniform routing $w_e = 1/E$ and moves away from it

only when the data support it. The fused representation and prediction are

$$z = \mathrm{LN}\Big(\sum_{e=1}^{E} w_e \, z_e\Big), \qquad \hat{y} = \mathrm{softmax}(W_c \, z + b_c), \tag{6}$$

with a single linear classifier $W_c \in \mathbb{R}^{7\times D}$. Unlike sparse top-$k$ MoE, all experts contribute to every sample. This is affordable with $E = 3$, and the routing stays differentiable without auxiliary noise. To prevent routing collapse onto one expert we add an importance-balancing regulariser

$$\mathcal{L}_{\mathrm{bal}} = E \sum_{e=1}^{E} \bar{w}_e^{\,2} - 1, \qquad \bar{w}_e = \frac{1}{B} \sum_{i=1}^{B} w_{ie}, \tag{7}$$

which is zero for uniform batch-level usage and still allows strong per-image specialisation.

3.5 Training objective

Weighted classification loss.

With class counts n_c in the training fold, $\alpha_c \propto n_c^{-1/2}$ (normalised to mean 1). The classification term is a soft-target weighted cross-entropy,

$$\mathcal{L}_{\mathrm{cls}} = -\sum_{c} \alpha_c \, q_c \log p_c. \tag{8}$$

Square-root weights are a milder form of inverse-frequency weighting. They were the best option in the legacy policy screen and cause fewer minority false positives (Section 2.1).

Supervised contrastive loss.

Re-weighting moves the decision boundaries but does not separate the classes in feature space. For the Pest/Fungi/Phytophthora cluster we therefore add the supervised contrastive loss of Khosla *et al.* [9] on a projection $h_i = g(z_i)/\|g(z_i)\|$, $g : \mathbb{R}^{512} \to \mathbb{R}^{128}$:

$$\mathcal{L}_{\mathrm{SC}} = \sum_{i \in I} \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(h_i^{\top} h_p / \tau)}{\sum_{a \in A(i)} \exp(h_i^{\top} h_a / \tau)}, \tag{9}$$

where $A(i)$ are all other samples in the batch, $P(i) = \{p \in A(i) : y_p = y_i\}$ and $\tau = 0.07$. Anchors without positives are skipped, and the term is applied only to non-MixUp batches, where labels are pure. It pulls images of the same class together and pushes the confused pairs apart.

Total loss.

$$\mathcal{L} = \mathcal{L}_{\mathrm{cls}} + \lambda_{\mathrm{SC}} \, \mathcal{L}_{\mathrm{SC}} + \lambda_{\mathrm{bal}} \, \mathcal{L}_{\mathrm{bal}}, \qquad \lambda_{\mathrm{SC}} = 0.1, \; \lambda_{\mathrm{bal}} = 0.01. \tag{10}$$

3.6 Two-stage refinement

Training three large pretrained experts on fewer than 2,500 images from random heads quickly distorts their features. We therefore use two stages.

Stage 1: alignment (5 epochs). All backbone parameters are frozen, including BatchNorm running statistics. Only the FPMs, router, classifier and contrastive head are trained (AdamW, $\eta_{\text{head}} = 3 \times 10^{-4}$, cosine decay). The randomly initialised layers learn to read the pretrained features before any gradient reaches the backbones.

Stage 2: refinement (up to 40 epochs). All parameters are unfrozen with discriminative learning rates, $\eta_{\text{backbone}} = 1 \times 10^{-5}$ and $\eta_{\text{head}} = 3 \times 10^{-4}$, a 2-epoch linear warm-up and cosine annealing to 1 % of the peak. Weight decay is 0.05 and gradients are clipped at norm 1. We use automatic mixed precision (bf16 where supported, otherwise fp16 with loss scaling), gradient checkpointing, a micro-batch of 8 with 4-step accumulation (effective batch 32) and an exponential moving average of the weights (decay 0.999).

Model selection. After every epoch we evaluate on the held-out fold with horizontal-flip test-time augmentation. We keep the checkpoint with the highest Macro-F1 and stop after 8 epochs without improvement in Stage 2. Because the selection uses the validation fold, the 5-fold CV estimate is selection-informed, as it was in the legacy campaign. We state this caveat with every result.

3.7 Evaluation protocol

For every fold k we compute accuracy, Macro-F1, weighted F1, Matthews correlation coefficient (MCC), per-class precision, recall and F1, and the confusion matrix. Tables report mean ± sample standard deviation over the five folds, and confusion matrices are pooled. The legacy models are evaluated on the same group-aware folds. Explanations are produced with routing-weighted Grad-CAM [10],

$$\mathrm{CAM}_e = \mathrm{ReLU}\Big(\sum_k \alpha_k^e P_e^k\Big), \quad \alpha_k^e = \frac{1}{h_e w_e} \sum_{u,v} \frac{\partial \hat{y}_c}{\partial P_e^k[u,v]}, \quad \mathrm{CAM} = \sum_e w_e \,\mathrm{up}(\mathrm{CAM}_e), \tag{11}$$

and summarised by the *lesion-focus ratio*, the share of CAM mass falling inside the foliage-and-lesion mask M.

4 Results

4.1 Setup

All experiments use the cleaned dataset and the fixed group-aware 5-fold split of Section 3.1. Training ran with PyTorch and timm under automatic mixed precision. Folds 0–3 were trained on an NVIDIA T4 (fp16) and fold 4 on an NVIDIA RTX 4090 (bf16). Data split, hyper-parameters, schedule and seed are identical; only data-loader workers and activation checkpointing, which affect speed but not the computation, were changed. Hyper-parameters are listed in Section 3.6 and in `configs/asmoe_v2.yaml` of the accompanying repository. Every number is the mean ± SD over five folds. Tables were generated automatically on 2026-11-06 12:15 UTC.

4.2 Main comparison

Table 4 Comparison with the legacy models under the same group-aware stratified 5-fold protocol (mean ± SD over five folds). Best Macro-F1 in bold.

Model	Res. (px)	Params (M)	Accuracy (%)	Macro-F1 (%)	MCC
EffNet-B0+MobileViT (ECA)†	224	10.5	83.55 ± 2.20	82.12 ± 2.67	0.801 ± 0.027
MobileViT-S†	224	4.9	84.92 ± 1.27	83.68 ± 1.42	0.817 ± 0.016
ASMOE V2 (ours)	512/480/384	160.7	92.63 ± 0.91	**91.75 ± 1.88**	0.910 ± 0.011

†Legacy model from the earlier campaign, re-evaluated on the group-aware folds. Res.: input resolution of each expert.

Table 4 compares ASMOE V2 with the two strongest legacy models, re-evaluated under the same leakage-free protocol. ASMOE V2 reaches 92.63 ± 0.91 % accuracy and 91.75 ± 1.88 % Macro-F1 (MCC 0.910), a change of +8.07 pp Macro-F1 and +7.71 pp accuracy over the legacy MobileViT-S baseline (83.68 ± 1.42 % Macro-F1). Pooled over the five validation folds, the number of errors falls from 464 (of 3,076 images) to 222 (of 3,012 cleaned images). ASMOE V2 does not reach the ≥ 95 % accuracy and Macro-F1 target set at the start of this work. The remaining gap is concentrated in Phytophthora and in the Fungi/Pest pair (Section 4.3). The improvement is larger than the fold-to-fold variation: the weakest ASMOE V2 fold (89.05 % Macro-F1, 91.36 % accuracy) is above the strongest legacy fold (85.15 % and 86.34 %), and the SD of accuracy falls from 1.27 to 0.91 pp. Because data cleaning, input resolution, training recipe and architecture all changed together, the gain reflects the complete pipeline rather than the mixture alone (Section 5).

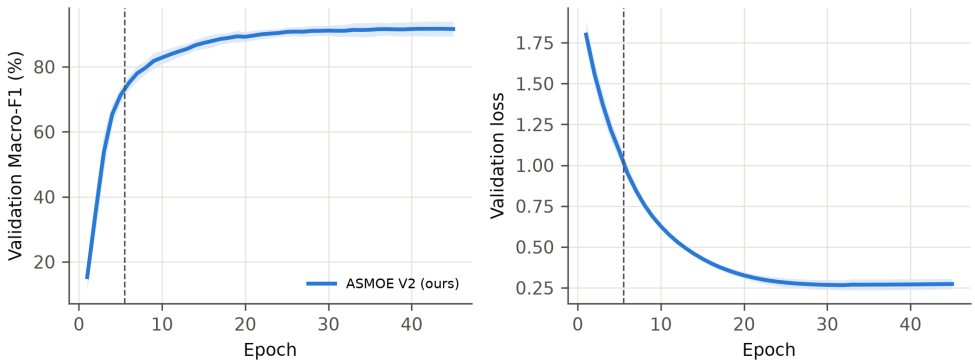


Fig. 4 ASMOE V2 validation Macro-F1 and loss per epoch (mean ± SD over five folds). The dashed line marks the end of the 5-epoch frozen-backbone stage. Compare with the legacy curves in Fig. 2.

Figure 4 shows that training converges: validation loss flattens after about 30 of the 45 epochs, and Macro-F1 changes by less than 1 pp over the last 15 epochs, so a longer schedule is unlikely to close the gap to the target.

4.3 Per-class recovery of minority classes

Table 5 Per-class F1 (%, mean ± SD over five folds). Best value per class in bold.

Class	MobileViT-S†	ASMOE V2	Δ (pp)
Bacteria	95.7 ± 2.3	**98.8 ± 0.8**	+3.1
Fungi	81.7 ± 3.1	**90.2 ± 1.4**	+8.5
Healthy	80.2 ± 3.5	**90.5 ± 4.0**	+10.3
Nematode	76.8 ± 4.0	**89.0 ± 8.1**	+12.1
Pest	78.4 ± 1.2	**91.6 ± 1.2**	+13.2
Phytophthora	85.6 ± 4.0	**87.0 ± 1.0**	+1.4
Virus	87.3 ± 3.3	**95.2 ± 1.2**	+7.9

†Strongest legacy model on the same group-aware folds. Δ: ASMOE V2 minus MobileViT-S.

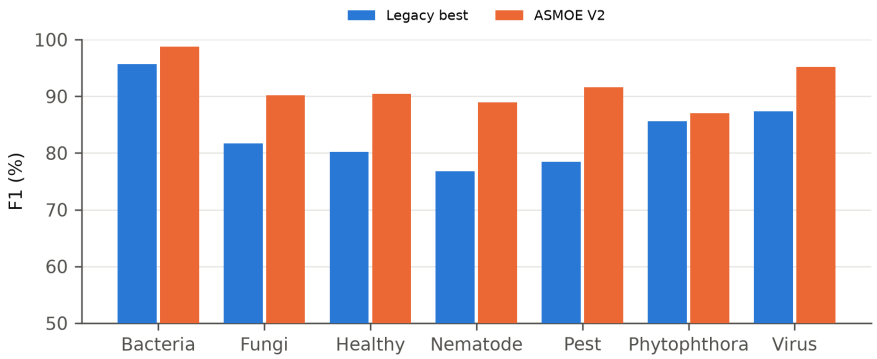


Fig. 5 Per-class F1 of the legacy baseline and ASMOE V2 (mean over 5 folds).

The legacy model's weakest classes were Nematode (76.8 % F1), Pest (78.4 %) and Healthy (80.2 %). ASMOE V2 reaches 89.0 % (+12.1 pp) on Nematode, 91.6 % (+13.2 pp) on Pest and 90.5 % (+10.3 pp) on Healthy (Table 5, Fig. 5). In the legacy model, Nematode and Healthy errors were mainly false positives caused by loss re-weighting (Section 2.1). The gain on these two classes comes mainly from higher precision. On the pooled matrices, Nematode precision rises from 67.0 to 82.3 % (false positives 30 → 14) and recall from 89.7 to 97.0 %. Healthy precision rises from 75.2 to 86.8 % (false positives 57 → 29) and recall from 86.1 to 94.5 %. Precision remains the limiting factor for both classes, so over-prediction of minority classes is reduced but not removed. The supervised contrastive term and the milder re-weighting were introduced together, so their individual contributions cannot be separated here. Removing the

six label-conflicting duplicates, all of which involved Pest, removes errors that no model could avoid, so part of the Pest gain is a measurement correction rather than a modelling improvement. Re-training the legacy MobileViT-S on the cleaned split with its original recipe would separate the two effects.

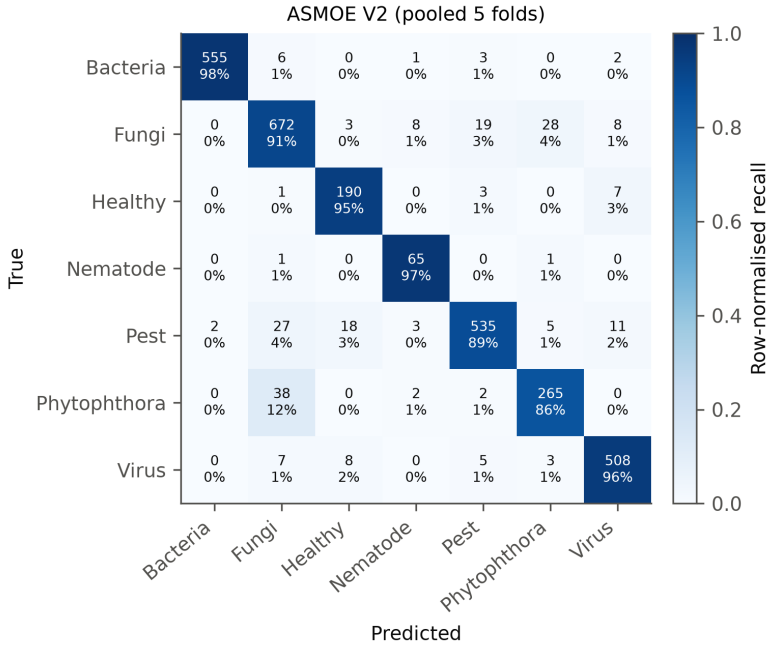


Fig. 6 Pooled 5-fold confusion matrix of ASMOE V2. Compare with Fig. 1.

Figure 6 should be read against the legacy matrix (Fig. 1). Pest→Fungi errors fall from 76 to 27 and Fungi→Pest errors from 69 to 19. Virus→Healthy errors fall from 20 to 8. Phytophthora→Fungi is the only major confusion that does not improve (31 to 38). Consistent with this, Phytophthora has the smallest F1 gain (+1.4 pp) and is now the weakest class (87.0 %). Closing the Phytophthora/Fungi gap is the most direct route to higher Macro-F1.

4.4 Routing behaviour

Figure 7 shows the average router output for each true class. If the experts specialise as intended, the global-attention expert should receive more weight for Virus and Healthy, where the evidence is diffuse, and the texture expert more weight for Pest and Fungi. The observed weights only partly follow this hypothesis. Swin-B receives its largest mean weights on Virus and Pest (0.45 each), but DenseNet-201 is used mainly for Bacteria (0.51) and receives only 0.17 on Pest and 0.27 on Fungi. EfficientNetV2-M dominates Healthy (0.50) and Nematode (0.52). The routing is therefore clearly class-dependent: the DenseNet-201 weight ranges from 0.09 (Nematode) to 0.51 (Bacteria),

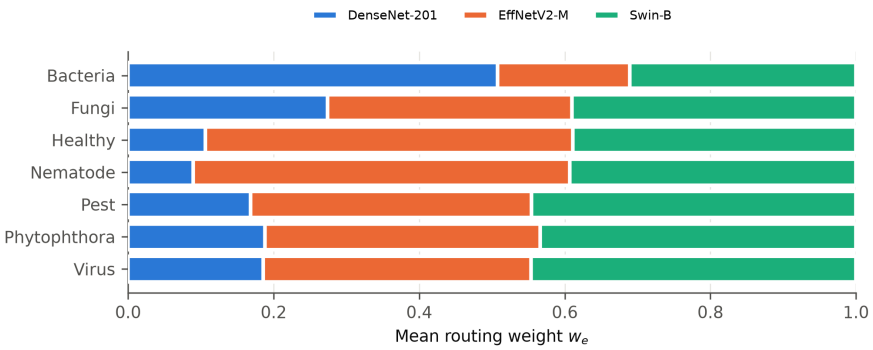


Fig. 7 Mean routing weight per expert, grouped by true class (validation folds).

and the fold-to-fold SD of every class-mean weight is at most 0.07. However, the learned division of labour is not the texture-versus-global split we expected. The routing is therefore input-dependent, but this study does not measure how much accuracy it adds over a fixed average of the same experts.

4.5 Explainability

Routing-weighted Grad-CAM (Fig. 8) shows which pixels each expert uses and how much each expert contributes. The expert maps differ in character. The EfficientNetV2-M maps are compact and centred on individual lesions or leaf regions. The DenseNet-201 maps cover larger leaf areas when the expert receives a high weight (Bacteria, $w = 0.67$) and are almost flat when its weight is low. The Swin-B maps are patchy and, in several examples (Pest, Virus), also highlight background regions at the image border. Because the fused map is the weight-averaged sum, these background responses carry through when Swin-B dominates the routing. The lesion-focus ratio (fraction of CAM mass inside the leaf mask, computed by `train.py gradcam`) quantifies this. On the eight fold-0 examples, a mean of 79 % of the fused CAM mass lies inside the leaf mask. Six of the eight examples exceed 0.68, and three exceed 0.93. The exception is a Pest image with a ratio of 0.31, where most attention falls outside the colour-based leaf mask. Eight images are too few for a general statement, but they indicate that the decision usually rests on foliage rather than soil or background.

5 Discussion

Evaluation protocol matters as much as architecture.

The legacy audit shows that the same MobileViT-S drops from 88.16% to 83.68% Macro-F1 when duplicate groups are kept within a single fold, although part of that gap is also due to a shorter training schedule. Results on this dataset that use random image-level splits, including our own legacy numbers and single-split results in the literature [2, 3], are therefore not directly comparable with the group-aware estimates in Table 4. We recommend that future work on this collection release its fold assignment or adopt the grouped split used here, which is published with the code.

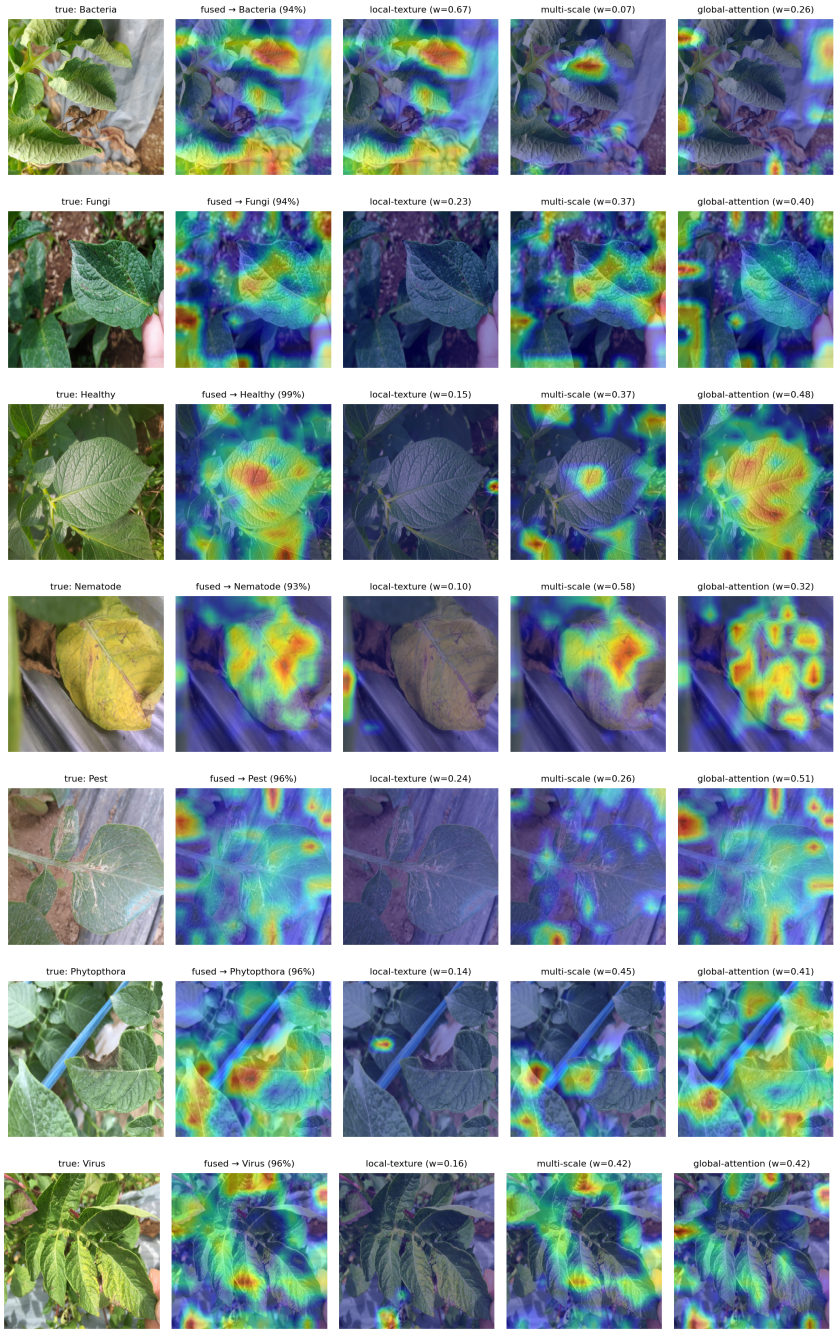


Fig. 8 Routing-weighted Grad-CAM for one correctly classified validation image per class (fold 0). Each row shows the input, the fused map with the predicted class and confidence, and the maps of the local-texture (DenseNet-201), multi-scale (EfficientNetV2-M) and global-attention (Swin-B) experts with their routing weights w_e for that image.

Where the mixture helps.

The largest F1 gains over the legacy model are on Pest (+13.2 pp), Nematode (+12.1 pp) and Healthy (+10.3 pp), and the Pest/Fungi confusion, the main error source of the legacy model, falls by about two thirds (145 to 46 pooled errors). Bacteria was already near ceiling, and Phytophthora barely improves (+1.4 pp) because its confusion with Fungi persists. The routing weights (Fig. 7) are strongly class-dependent but do not follow the intended texture-versus-global split, so the gains cannot yet be attributed to expert specialisation. Part of the improvement also comes from data cleaning, higher resolution and the new training recipe, which changed together with the architecture.

Limitations.

- The reported gain is that of the complete pipeline; quantifying the contribution of each component is left for future work.
- Model selection uses the validation fold (early stopping), so the cross-validated estimate is selection-informed. An external test set from another region or season would give an unbiased estimate.
- Each validation fold contains only 11–17 Nematode images, so Nematode F1 has a standard error of several percentage points. Differences below about 5 pp on this class should not be over-interpreted.
- The ROI mask is a colour heuristic. Promptable segmentation models may give cleaner crops on cluttered backgrounds.
- ASMOE V2 has about 160 M parameters, against 4.9–10.5 M for the legacy models. Knowledge distillation into a compact student is needed for on-device use.

6 Conclusion

We audited an earlier experimental campaign on the Potato Leaf Disease Dataset in Uncontrolled Environment and identified label-conflicting duplicates, near-duplicate leakage, re-weighting-induced false positives, lesion-destroying augmentation and under-training as the main bottlenecks. To address them we introduced a leakage-free protocol and ASMOE V2, a heterogeneous adaptive soft mixture of experts trained with two-stage refinement and supervised contrastive learning. Under group-aware 5-fold cross-validation, ASMOE V2 reaches 91.75% Macro-F1 and 92.63% accuracy, compared with 83.68% and 84.92% for the strongest legacy model. Future work will test the model on externally collected fields, distil the mixture into a mobile-sized network, and replace the colour-based ROI with learned leaf segmentation.

Declarations

Funding. [To be completed by the authors.]

Competing interests. The authors declare no competing interests. [Confirm.]

Ethics approval and consent to participate. Not applicable.

Consent for publication. Not applicable.

Data availability. The images are publicly available as the Potato Leaf Disease Dataset in Uncontrolled Environment [1]. The cleaned file list, duplicate groups and fold assignment used in this study are released with the code.

Code availability. Code, configurations, per-fold results and the scripts that generate every table and figure in this article are available at https://github.com/beyondmme/asmoe-potato-v2.

Author contribution. T.L.V. (lead author): conceptualisation, methodology, software, investigation, formal analysis, visualisation, project administration, writing – original draft. N.H.A.K.: data curation, validation, investigation, writing – review and editing. N.N.T.N.: validation, visualisation, writing – review and editing. All authors read and approved the final manuscript.

References

[1] Shabrina, N.H., Indarti, S., Maharani, R., Kristiyanti, D.A., Irmawati, Prastomo, N., Adilah, M.T.: A novel dataset of potato leaf disease in uncontrolled environment. Data in Brief **52**, 109955 (2024) https://doi.org/10.1016/j.dib.2023.109955

[2] Saleh, H., McCann, M., Creaven, R., Breslin, J.G., El-Sappagh, S.: Hybrid CNN-Swin Framework for Detection and Classification of Potato Leaf Diseases. Conference manuscript, Insight Research Ireland Centre for Data Analytics, University of Galway (2025)

[3] Hoang, P.-H., Phan, T.-T.-H.: Cross-architectural mixture-of-experts with adaptive soft routing for plant leaf disease classification. Array **32**, 101261 (2026) https://doi.org/10.1016/j.array.2026.101261

[4] Yun, S., Han, D., Oh, S.J., Chun, S., Choe, J., Yoo, Y.: Cutmix: Regularization strategy to train strong classifiers with localizable features. In: IEEE/CVF International Conference on Computer Vision (2019)

[5] Zhang, H., Cisse, M., Dauphin, Y.N., Lopez-Paz, D.: mixup: Beyond empirical risk minimization. In: International Conference on Learning Representations (2018)

[6] Huang, G., Liu, Z., Maaten, L., Weinberger, K.Q.: Densely connected convolutional networks. In: IEEE Conference on Computer Vision and Pattern Recognition (2017)

[7] Tan, M., Le, Q.: Efficientnetv2: Smaller models and faster training. In: International Conference on Machine Learning (2021)

[8] Liu, Z., Lin, Y., Cao, Y., Hu, H., Wei, Y., Zhang, Z., Lin, S., Guo, B.: Swin transformer: Hierarchical vision transformer using shifted windows. In: IEEE/CVF International Conference on Computer Vision (2021)

[9] Khosla, P., Teterwak, P., Wang, C., Sarna, A., Tian, Y., Isola, P., Maschinot, A., Liu, C., Krishnan, D.: Supervised contrastive learning. In: Advances in Neural Information Processing Systems (2020)

[10] Selvaraju, R.R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., Batra, D.: Grad-cam: Visual explanations from deep networks via gradient-based localization. In: IEEE International Conference on Computer Vision (2017)